

CONTACT	Email: vivwhite@cs.ubc.ca Website: vivianwhite.github.io LinkedIn: linkedin.com/in/vivian-white
RESEARCH INTERESTS	I am passionate about furthering our scientific understanding of black-box deep networks to develop robust, interpretable, and efficient models for computer vision. I currently study (1) interpretability and (2) energy efficiency of deep models during test-time adaptation.
PUBLICATIONS	3. V White, M Chaudhary, G Wolf, G Lajoie, KD Harris. Learning Stochastic Rainbow Networks. NeurIPS Workshop on Scientific Methods for Understanding Deep Learning (Sci4DL), 2024. 2. V White, A White, J Wild, T Nguyen, F Huang. Human Error Scenario Analysis of Software Defects. ISSRE Workshop on Human Factors for Software Dependability, 2024. 1. V White, M Chaudhary, G Wolf, G Lajoie, KD Harris. Learning and Aligning Structured Random Feature Networks. ICLR Workshop on Representational Alignment, 2024.
EDUCATION	The University of British Columbia (UBC), Vancouver, British Columbia, CA. · Ph.D., Computer Science, expected 2030 · Certificate in Instructional Skills for Graduate Students Western Washington University (WWU), Bellingham, Washington, USA. · M.S., Computer Science, June 2025, GPA 4.00/4.00 · B.S., Computer Science, June 2024, GPA 3.78/4.00 · Minors in Mathematics and Honors Interdisciplinary Studies
GRANTS AND AWARDS	Graduate · Maria Klawe Service Award, UBC CS Dept., 2026 · AML-TN Funded Trainee, UBC CS Dept., 2025 · CS Merit Scholarship, UBC CS Dept., 2025 · Outstanding MS Graduate in CS, WWU, 2025 · Dean J. Alan Ross Travel Fund Award, WWU, 2024 · CS Graduate Fellowship, WWU CS Dept., 2024 · Graduate Recruitment Tuition Waiver, WWU, 2024 Undergraduate · Outstanding BS Graduate in CS, WWU, 2024 · 1st-place for CS poster presentation, Emerging Researchers National (ERN) Conference, 2024 · Barbara Ellen Maguire-Veith Family Scholarship, WWU CS Dept., 2023 · Fellow, International Network for Bio-Inspired Computing (IN-BIC), 2023 · Lars and Elaine Giusti Scholarship for CS, WWU CS Dept., 2022 · CS/Math Distinguished Scholar Award, WWU CS Dept., 2020-2023 · Western Foundation Distinguished Scholar Award, WWU, 2020 · Merit Scholarship, WWU, 2020 · Admissions Achievement Award, WWU, 2020 · Admissions Annual Scholarship, WWU, 2020 · Scholarship Award, WSECU, 2020
EXPERIENCE	Graduate Research Assistant, UBC and Vector Institute for AI, Sept 2025–present · Studying interpretability and efficiency of test-time adaptation with Dr. Evan Shelhamer · Supported by UBC's Advanced Machine Learning – Training Network (AML-TN) trainee award Masters Intern, Pacific Northwest National Laboratory (PNNL), June–Aug 2025 · Worked on Math for AI Assurance with Dr. Henry Kvinge · Fine-tuned a base LLM using Group Relative Policy Optimization for statistical weight analysis Graduate Teaching Assistant, WWU, Dec 2024–June 2025 · Taught weekly labs and graded for an introductory data science class · Graded for a graduate analysis of algorithms class Graduate Research Assistant, WWU, June 2024–June 2025

- Developed stochastic rainbow networks with Drs. Kameron Harris, Guy Wolf, and Guillaume Lajoie

Undergraduate Research Assistant, WWU, June 2022–June 2024

- Developed randomized scattering convolutional networks and learnable structured random features with Drs. Kameron Harris, Guy Wolf, and Guillaume Lajoie

Research Intern, Mila - the Quebec AI Institute, Montréal, QC, July–Sept 2023

- Co-advised by Drs. Guy Wolf and Guillaume Lajoie studying learnable structured random features
- Funded by IN-BIC fellowship grant through National Science Foundation (NSF) AccelNet program

Undergraduate Research Assistant, WWU, April 2021–March 2022

- Worked with Dr. Scott Wehrwein on project applying machine learning techniques to international border legibility tasks for computer vision

PRESENTATIONS

Green TTA: Benchmarking the Energy Efficiency of Test-Time Adaptation

- AI/CRV, Burnaby, BC, May 2026 (poster)

What Changed? Interpreting Model Updates During Training and Test-Time Adaptation

- Canadian Celebration of Women in Computing, Vancouver, BC, Nov 2025 (oral)
- ACM Celebration of Cascadia Women in Computing, Vancouver, BC, Oct 2025 (poster)

Analysis of LLM Weights During RL Reasoning Finetuning

- Gold Intern Symposium, PNPL, virtual, Aug 2025 (oral)

Learning Stochastic Rainbow Networks

- Graduate Research Symposium, WWU, Bellingham, WA, June 2025 (oral)
- ERN Conference, Atlanta, GA, Mar 2025 (oral)
- NeurIPS Workshop on the Science of Deep Learning, Vancouver, Canada, Dec 2024 (poster)

Learning and Aligning Structured Random Feature Networks

- NeuroAI Workshop, University of Oregon, Eugene, OR, Aug 2024 (oral)
- CoNectome Symposium, University of Washington, Seattle, WA, May 2024 (poster)
- ICLR Workshop on Representational Alignment, Vienna, Austria, May 2024 (poster)
- ERN Conference, Washington DC, Mar 2024 (poster)

Randomized Scattering Convolutional Networks

- DeepMath Conference, Johns Hopkins University, Baltimore, MD, Nov 2023 (poster)
- NeuroAI Workshop, Mila, Montréal, QC, Oct 2023 (poster)
- NSF S-STEM Scholars Meeting, Washington DC, Sept 2023 (poster)
- NeuroAI Workshop, University of Washington, Seattle, WA, Sept 2022 (poster)

SERVICE

- Reviewer: NeurIPS 2025, Sci4DL Workshop @ ICLR 2026 (Outstanding Reviewer), TTU Workshop @ ICLR 2026
- VP of Academics, UBC CSGSA, 2026-2027
- Co-Organizer, AI/CRV Workshop on AI for Sustainability and Science, 2026
- Co-Founder, UBC Graduate Women in Computer Science (GWICS), 2025-

TECHNICAL SKILLS

Languages: Python (PyTorch, HuggingFace, numpy, pandas, scikit-learn), Java, C, Javascript

Systems: Bash, Linux, L^AT_EX, git, HPC cluster environments, AWS

SOFT SKILLS

Event planning, leadership, communication, community building

REFERENCES

Evan Shelhamer

Assistant Professor, Department of Computer Science, The University of British Columbia
CIFAR AI Chair, Vector Institute for AI

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Kameron Decker Harris

Associate Professor, Department of Computer Science, Western Washington University

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